Chamarthy Praneeth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.53
Bhashyam Junior College	Class XII (Andhra Pradesh Board)	2023	97.00%
Sri Chaitanya School	Class X (CBSE)	2021	93.40%

EXPERIENCE

Nutanix
Technical Staff Intern (MTS)

- **Designed** an end-to-end UI test automation framework for Nutanix's Projects 2.0 platform using Python and Playwright, structured as a 3-tier architecture (Page Objects, Operations, Scenarios) to cut flaky tests by 40%.
- **Developed** Page Object Models with centralized locators and synchronous/asynchronous methods, automating 60+ test cases and decoupling UI logic for resilient front-end changes.
- **Devised** parallel multi-browser scenario runners with cumulative reporting, automatic screenshot/video capture on failure, and deterministic locator-state assertions, lifting CI reliability by 30%.
- **Accelerated** developer feedback by integrating the regression suite into CI/CD, reducing suite execution time by 45% while supporting 50 concurrent users.

Research Assistant
Prof. Sathya Peri, IIT Hyderabad

- **Investigated** lock-free and wait-free synchronization by designing concurrent shared-array algorithms in C++20 and benchmarking their throughput and p99 latency under high contention.

PROJECTS

TokenWise - Knowledge-Graph Driven Context Retrieval for AI Coding

- **Pioneered** a knowledge-graph-driven context-retrieval system for AI coding assistants, integrated with OpenCode and Nutanix NAI, projecting ~$700K in annual savings across 3,500 users.
- **Created** a Knowledge Graph MCP server using TypeScript, tree-sitter, and Neo4j that scans codebases into a persistent graph, replacing 20+ file reads per query and reducing input tokens by 50%.
- **Introduced** semantic routing to classify query intent, directing simple queries to lightweight models and complex reasoning to frontier models via Nutanix NAI and FAISS, saving ~7,000 tokens per request.

ProbScript - Probabilistic Programming Language Compiler | *Source Code*

- **Authored** a C++17 compiler frontend using Flex and LALR(1) Bison that generates a typed AST with line- and column-precise source-located error reporting.
- **Defined** a probabilistic type system with 3 first-class types (Distribution, RandomVariable, Event) plus semantic analysis for type checking, inference, scope, and symbol resolution.
- **Validated** correctness across 150+ test cases, including 20 negative tests across 8 categories, using a golden-output methodology with a 100% pass rate.

Multi-Write - Lock-Free & Wait-Free Concurrent Shared Array | *Research · Prof. Sathya Peri*

- **Investigated** lock-based vs. non-blocking synchronization by building four thread-safe shared-array designs in C++20: global lock, fine-grained lock, lock-free snapshot CAS, and wait-free.
- **Achieved** wait-free progress via per-thread announcement slots and a propose-and-help protocol that bounds retries, reducing p99 multi-write tail latency under high contention.
- **Conducted** benchmark sweeps across read-heavy, mixed, and multi-write-only workloads, measuring throughput, p99 latency, and peak memory with correctness verified through concurrent stress tests and ThreadSanitizer.

SKILLS

- **Programming Languages:** C, C++, JavaScript, Bash
- **Web Technologies:** HTML, CSSReact, Next.js, Tailwind CSS, Node.js, Express.js
- **Databases:** MySQL, MongoDB
- **Other Tools:** Git, GitHub, LaTeX
- **DevOps:** Docker, Vercel, Netlify, Linux, Bash Scripting
- **Competitive Programming:** Solved 200+ problems on LeetCode (handle: spamprx)

POSITIONS OF RESPONSIBILITY

Media Council, IIT Hyderabad | *Web Head*

- **Spearheaded** development of the official Students' Gymkhana portal for 3,000+ students using React and Node.js, boosting engagement by 35% and migrating the platform to IITH subdomain architecture with 5+ council teams.

Milan, Prayas & Diesta Web Teams | *Coordinator*

- **Launched** accessible, mobile-first web platforms with real-time scoreboards for 3 major campus events, processing 1,000+ registrations and reducing page load times by 50%.

RELEVANT COURSES

- **Mathematics & Statistics:** Calculus, Discrete Mathematics, Probability and Random Variables, Introduction to Statistics, Linear Algebra, Transformation Techniques
- **Theory & Algorithms:** Data Structures, Algorithms, Theory of Computation, Compilers
- **Systems & Networks:** Operating Systems, Computer Architecture, Database Management Systems, Networks, Software Development Fundamentals
- **Specialized:** Machine Learning, Software Engineering